**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 019 · CPU Performance

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the raw bitmap size in bits before metadata or compression?

Q2. For cost = price × quantity, which is an input?
A. Calculated total cost
B. Printed receipt
C. Quantity
D. Displayed bill

Q3. Why use pseudocode when designing an algorithm?

Q4. A flowchart must accept ages 13–18 and reject others. “A diamond only shows calculations with no conditions.” Correct this. How should it check age?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Clock speed — The number of clock cycles the CPU carries out per second, measured in hertz.

Hertz (Hz) — Unit of clock speed: one clock cycle per second.

Cache — Small, fast memory that stores frequently used data and instructions for the CPU.

CPU core — Part of a CPU that processes instructions.

Parallel processing — Processing more than one task at the same time using multiple CPU cores.

## New learning

Clock speed: Clock speed is measured in hertz (Hz). A higher clock speed means the CPU carries out more clock cycles each second, so it can generally process instructions faster.

Cache size: Cache is small, fast memory used to store frequently used data and instructions. A larger cache can reduce the need to fetch them from slower main memory (RAM), improving performance.

Number of cores: A core processes instructions. A processor with more cores can process more tasks at the same time, which can improve performance when several programs are running.

Compare before you decide: Look at all three factors: clock speed, cache size and number of cores. For example, one CPU may have a higher clock speed while another has more cores and a larger cache. Explain which factors matter for the stated task.

## Worked example

Using the CPU A and CPU B specifications shown, explain TWO reasons CPU B may be better for running several programs at the same time. [4 marks]

1. Read the specifications shown: CPU A has the higher clock speed. CPU B has more cores and a larger cache. The task involves running several programs.

2. First reason: CPU B has four cores rather than two [1]. More cores can process more tasks at the same time, improving multitasking [1].

3. Second reason: CPU B has 8 MB cache rather than 4 MB [1]. More frequently used data and instructions can be accessed quickly, reducing slower RAM access [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

4. Check the command word 'explain': each reason must link a CPU characteristic to an improvement. CPU A's higher clock speed is a valid comparison, but we cannot guarantee which CPU is faster in every task.

## We Do · Practise with guidance

**Guided 1**

Compare CPU C and CPU D

Worked method: CPU C: 3.6 GHz, 2 cores, 4 MB cache. CPU D: 3.0 GHz, 6 cores, 12 MB cache. A pupil runs several programs at the same time. Explain TWO reasons why CPU D may be a better choice. [4 marks]

**Guided 2**

Clock speed

Worked method: CPU E runs at 3 GHz and CPU F at 2 GHz. Identify which has the higher clock speed and explain one possible performance advantage. [2 marks]

**Guided 3**

Cache size

Worked method: A processor is upgraded from 2 MB of cache to 8 MB. Explain how this could improve CPU performance. [2 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Predict the message printed.

```
clockA = 3
clockB = 2
if clockA > clockB then
    print("A has higher clock speed")
endif
```

predict

## Question 1

State the three characteristics of a CPU that affect performance. [3 marks] Source: Original J277-style question based on OCR J277 1.1.2.

## Question 2

A college chooses a CPU for computers running several applications. CPU G: 3.9 GHz, 2 cores, 6 MB cache. CPU H: 3.1 GHz, 8 cores, 16 MB cache. Explain TWO reasons why CPU H could be better for this use. [4 marks] Source: Original J277-style application question based on OCR J277 1.1.2.

## Question 3

Explain how clock speed, cache size and the number of cores each affect CPU performance. [6 marks] Source: Adapted from your Paper 1 revision booklet, section 1.1.2.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Exit Ticket · Answer before checking

A processor with more cores can improve performance when several programs are running.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Width × height × bits per pixel.

Q2. C — Quantity

Q3. Pseudocode shows steps in order without needing one programming language's exact syntax.

Q4. A diamond is a decision. Its branches follow different conditions. Check that each input follows the right path.

## I Do / We Do pages · Guided practice

Q1. CPU C: 3.6 GHz, 2 cores, 4 MB cache. CPU D: 3.0 GHz, 6 cores, 12 MB cache. A pupil runs several programs at the same time. Explain TWO reasons why CPU D may be a better choice. [4 marks]

Q2. CPU E runs at 3 GHz and CPU F at 2 GHz. Identify which has the higher clock speed and explain one possible performance advantage. [2 marks]

Q3. A processor is upgraded from 2 MB of cache to 8 MB. Explain how this could improve CPU performance. [2 marks]

## You Do pages · Code First

```
clockA = 3
clockB = 2
if clockA > clockB then
    print("A has higher clock speed")
endif
```

Expected result: A has higher clock speed

## You Do pages · Independent questions

Q1. Clock speed [1]; cache size [1]; number of CPU cores [1].

Q2. CPU H has eight cores rather than two [1], so it can process more tasks at the same time [1]. CPU H has 16 MB cache rather than 6 MB [1], so more frequently used instructions/data can be accessed quickly, reducing slower RAM accesses [1]. CPU G has the higher clock speed; the best choice depends on the workload.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Original practice marking points, not an official OCR mark scheme.

Q3. Clock speed: a higher clock speed provides more clock cycles each second [1], allowing the CPU to process instructions faster [1]. Cache size: a larger cache can store more frequently used data/instructions [1], reducing access to slower main memory [1]. Number of cores: more cores can process more tasks at the same time [1], improving multitasking/performance for tasks that can run in parallel [1]. This point-by-point allocation is a teaching model; assess extended answers against the relevant question criteria.

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**